



Attention Lecture 3

Alex Holcombe

2024-05-09





TODAY

- Understanding the hand-wave and monster onset
- Top-down vs. bottom-up attention (Ch. 5)
- Parallel feature processing (Ch. 5, Ch. 7)
- Absence of feature pairing (Ch. 7)
- Feature integration theory (Ch. 9)
- It feels like we see everything at once (Ch. 3.5)
- Distraction experiment



Lecture 3 outline

3. Treisman's diagram and combining features

- Individual-feature salience is computed in parallel
- Flat search slopes (parallel) vs. steep (serial)
- Combining features from two different objects takes the longest, but even within an object takes awhile (both require focused attention)
- It feels like we see everything at once
- The mysterious non-selective pathway



Understanding the hand-wave and monster onset

- My hand motion triggered a shift of attention to it. How?
 - Visual cortex signals the location of odd motions
 - Bottom-up attention encourages a shift of attention
 - Your attention may leave my hand quickly, because it's not interesting
- The monster appeared on the screen.
 - Visual cortex signalled the location of a sudden onset (Ch. 6)
 - Bottom-up attention encourages a shift of attention
 - Attention may linger, because the monster is interesting (top-down attention)



Bottom-up attention

1. The monster appeared on the screen
 1. Visual cortex signalled the location of a sudden onset (Ch. 6)
 2. Bottom-up attention encourages a shift of attention
 3. *Only then* is the monster routed through the bottleneck and recognized to be a monster
 4. Attention may linger there, because monsters are interesting / arousing
- Steps 2-3 happen faster than you can notice!



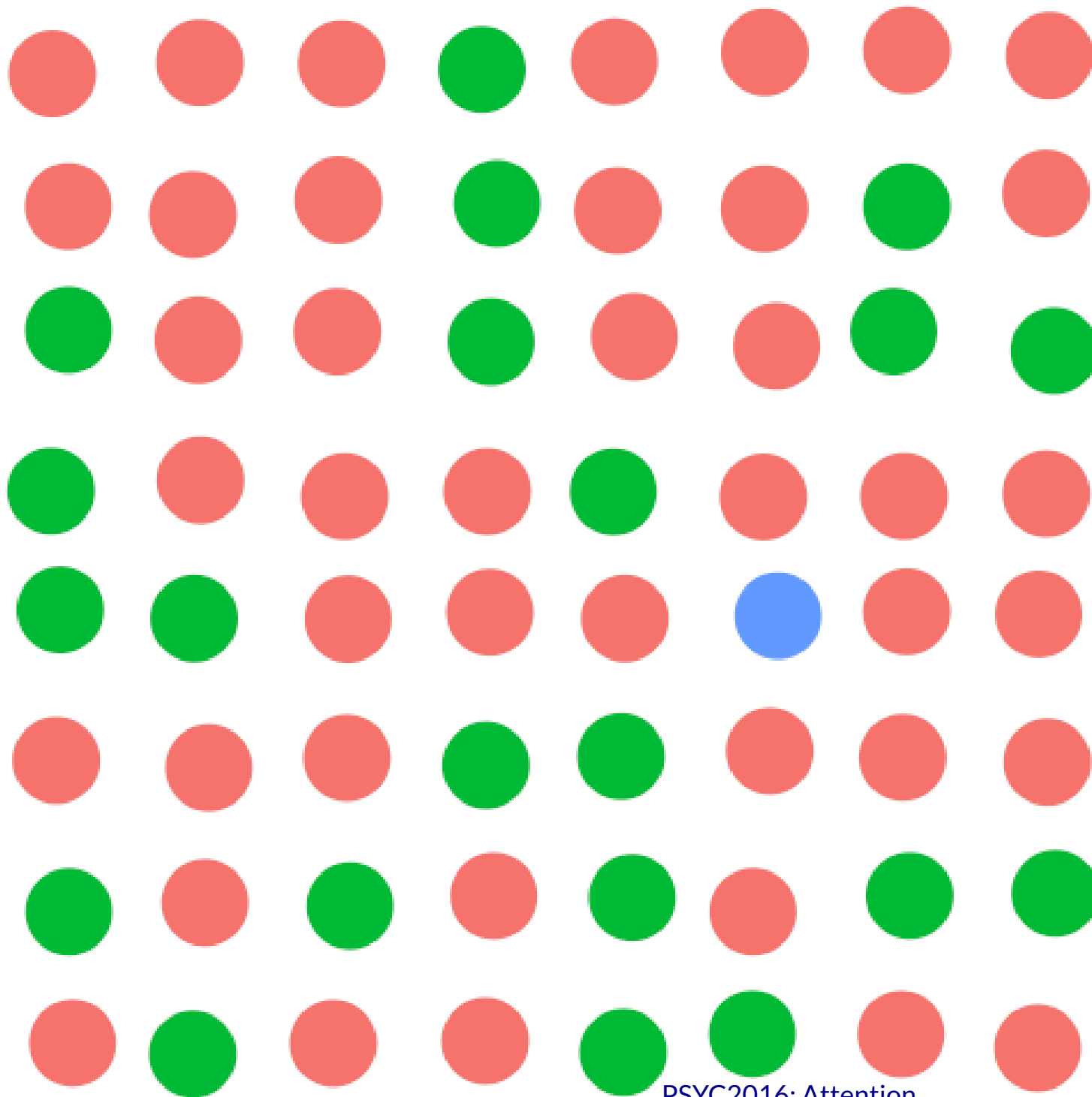
Did attention go to the monster because it was a monster? NO!

- The monster appeared on the screen
 - Visual cortex signalled the location of a sudden onset (Ch. 6)
 - Bottom-up attention encourages a shift of attention
 - *Only then* is the monster routed through the bottleneck and recognized to be a monster.
 - Attention may linger there, because monsters are interesting / arousing



Parallel feature processing (Ch. 5 and 7)



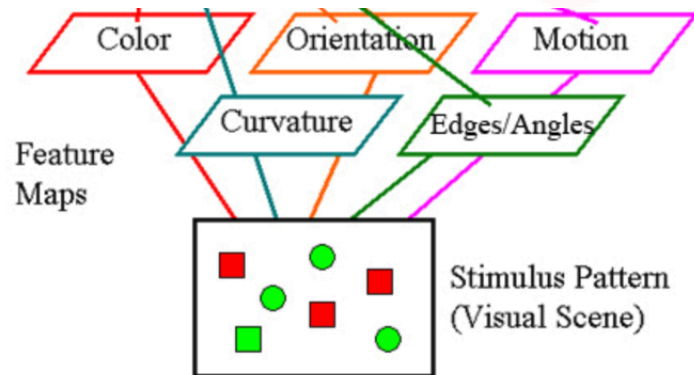


Parallel feature processing

- Visual cortex detects color difference
 - Processes whole visual field simultaneously
 - Oddball color signal propagates (saliency)
 - Drives bottom-up attention



Parallel feature processing



Pre-attentive

- automatic
- pre-conscious
- parallel

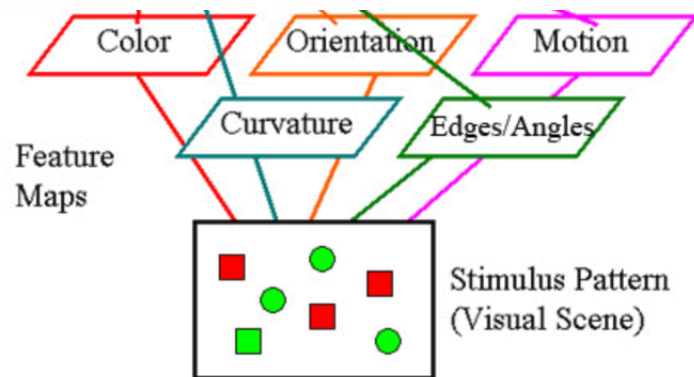
Proximal Stimulus

PSYC2016: Attention



Features

- Feature *dimensions* include color, motion, orientation
 - Feature *values*: red, blue for color
 - Feature *values*: leftward, downward for motion



Pre-attentive

- automatic
- pre-conscious
- parallel

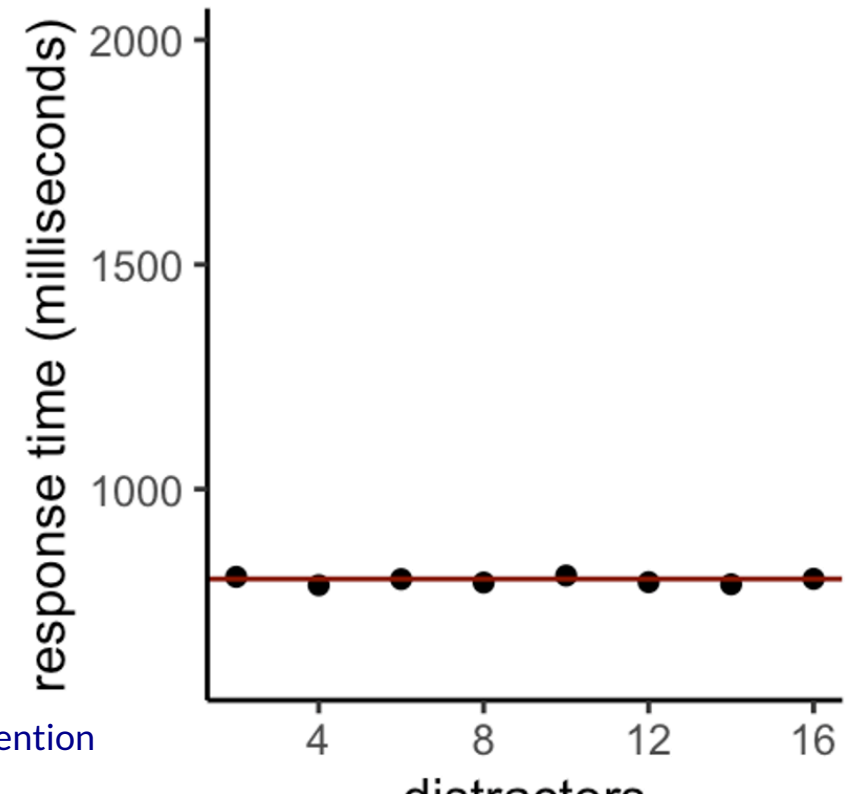
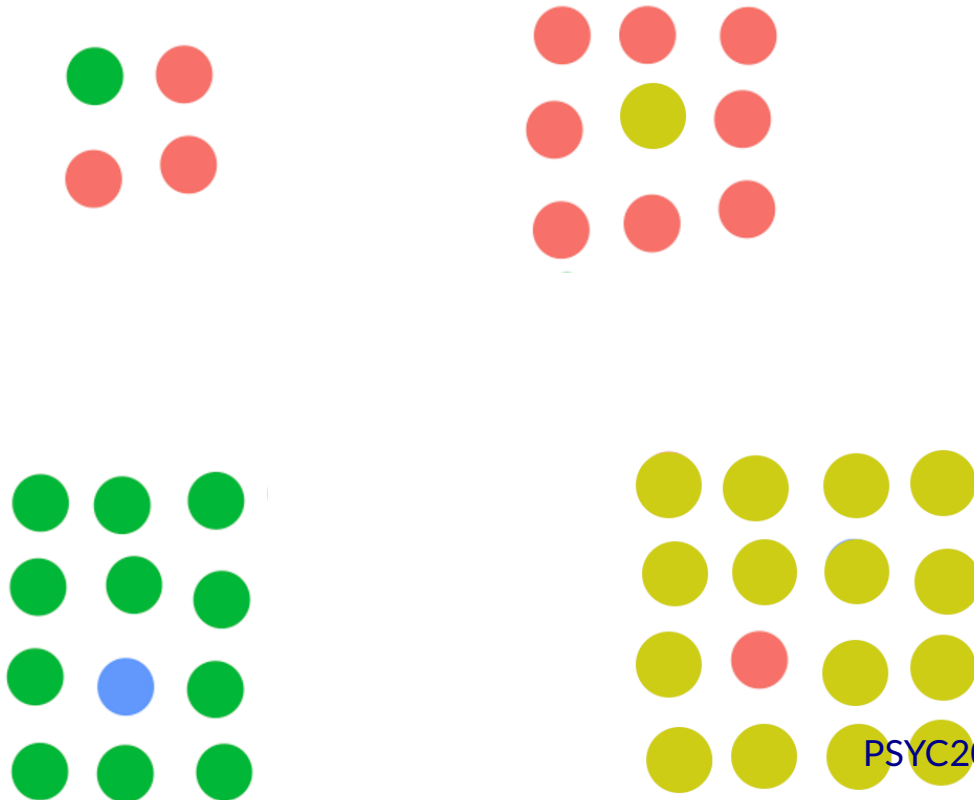
Proximal Stimulus

PSYC2016: Attention



Parallel processing

Odd feature values are salient



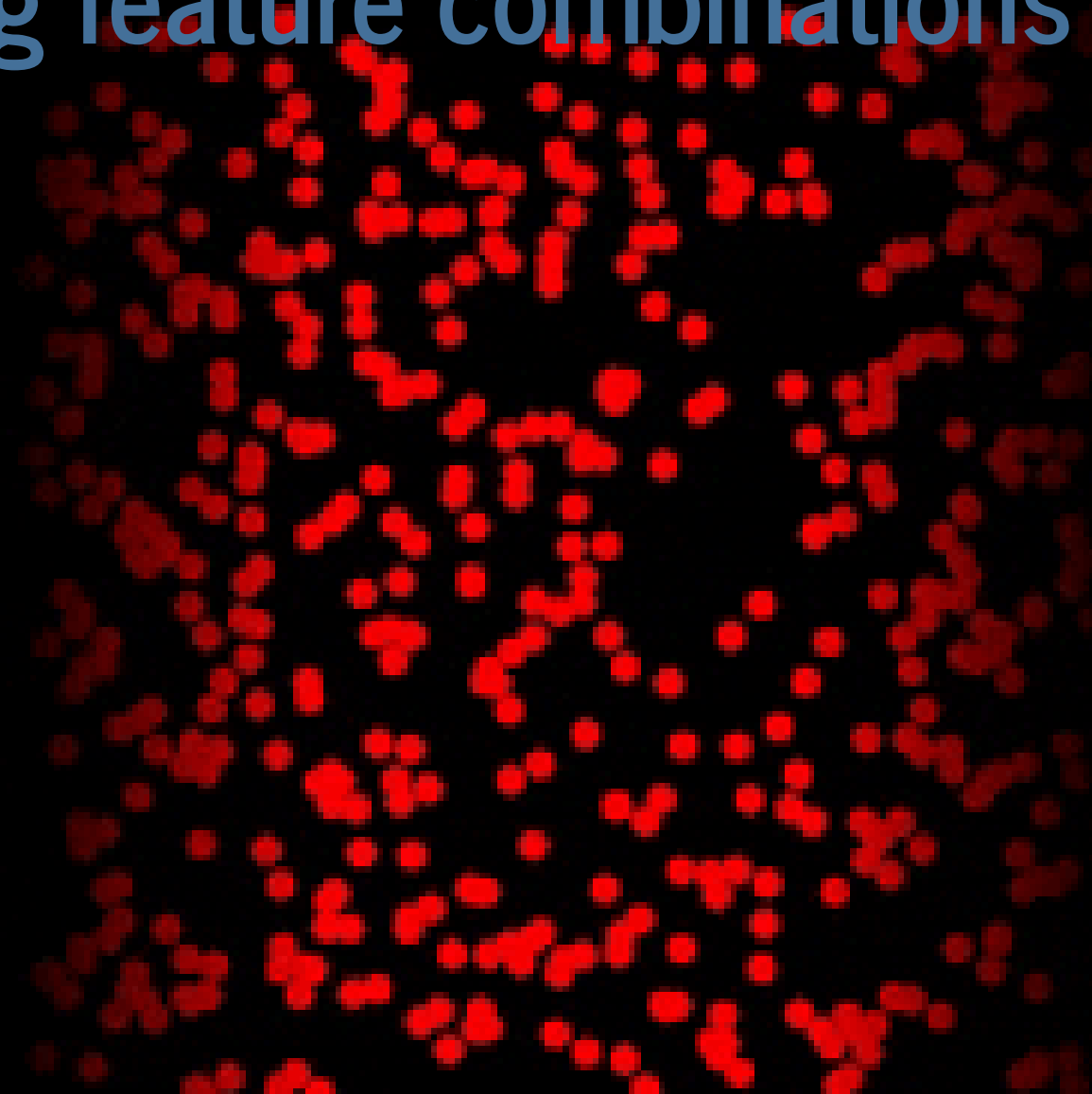
Feature processing

- *Within* a feature dimension, salience computation is massively parallel
- Identifying feature *combinations* is slow.



Identifying feature combinations is slow

Chapter 8



Identifying feature combinations is slow

Possibly doesn't occur unless selective attention routes the features through the bottleneck.





From lecture 2, Didn't understand: Where

The different brain regions responsible for object and feature selection

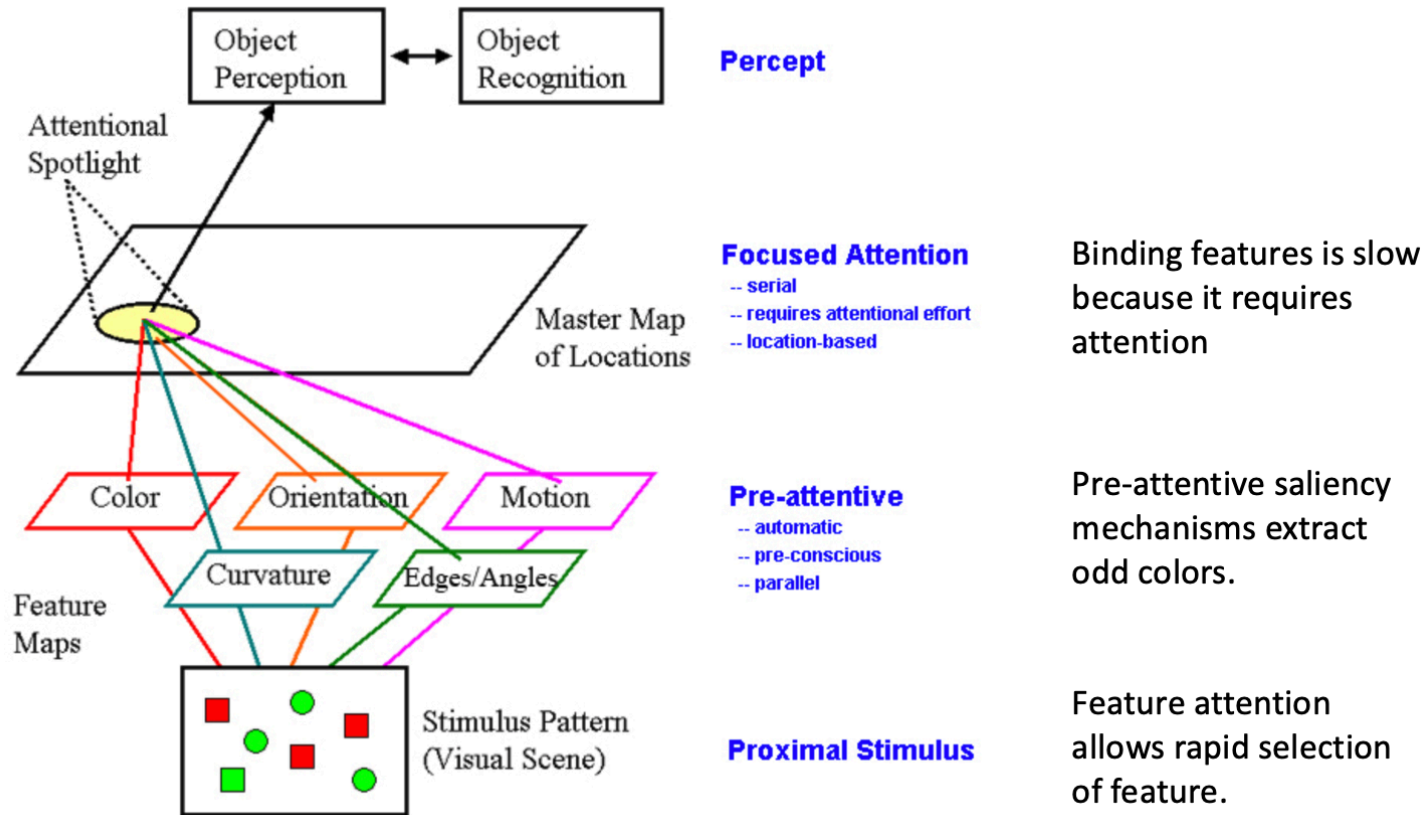
What you need to know:

- dorsal = parietal = where = location processing
- ventral = temporal = what = object recognition

What



Feature integration theory of Treisman



Binding features is slow because it requires attention

Pre-attentive saliency mechanisms extract odd colors.

Feature attention allows rapid selection of feature.

- *Within* a feature dimension, salience computation is massively parallel
- Identifying feature combinations is slow

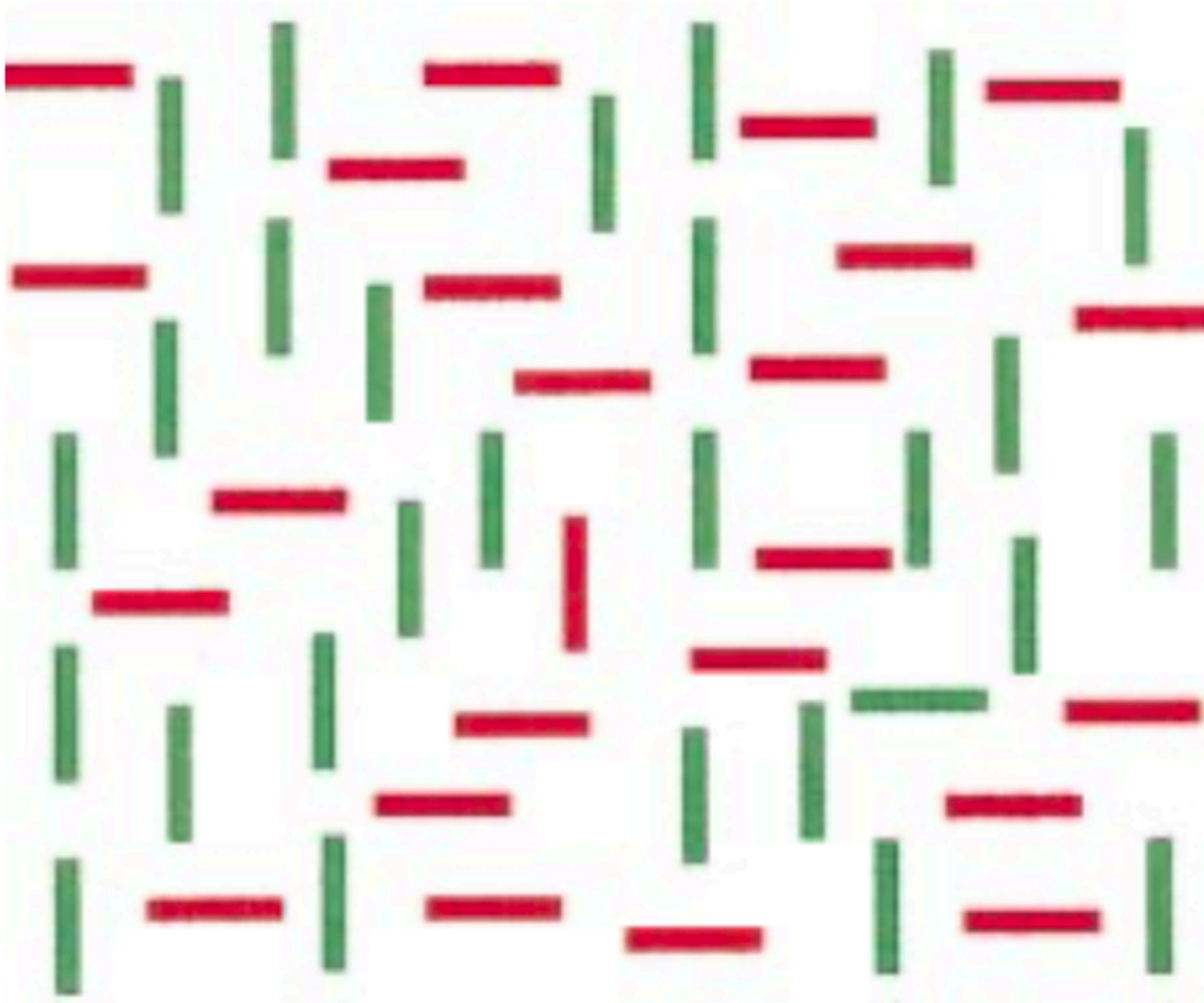




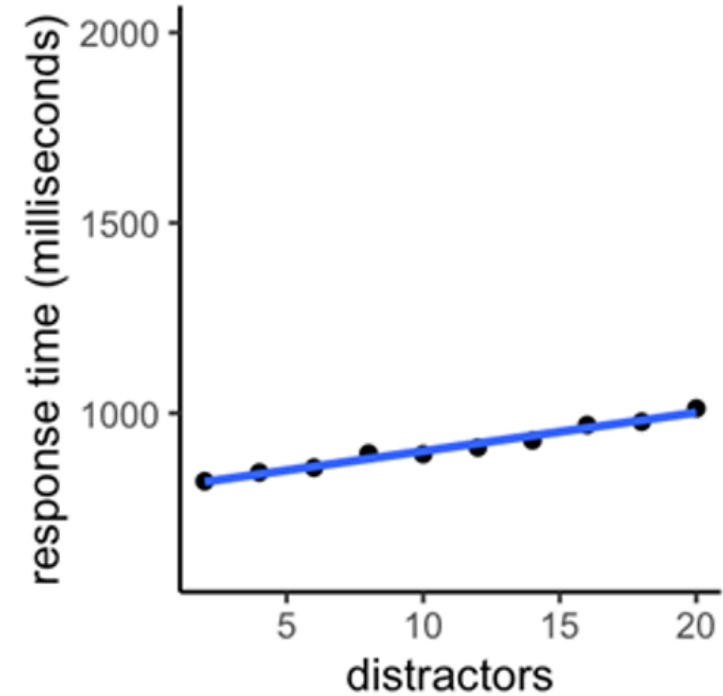
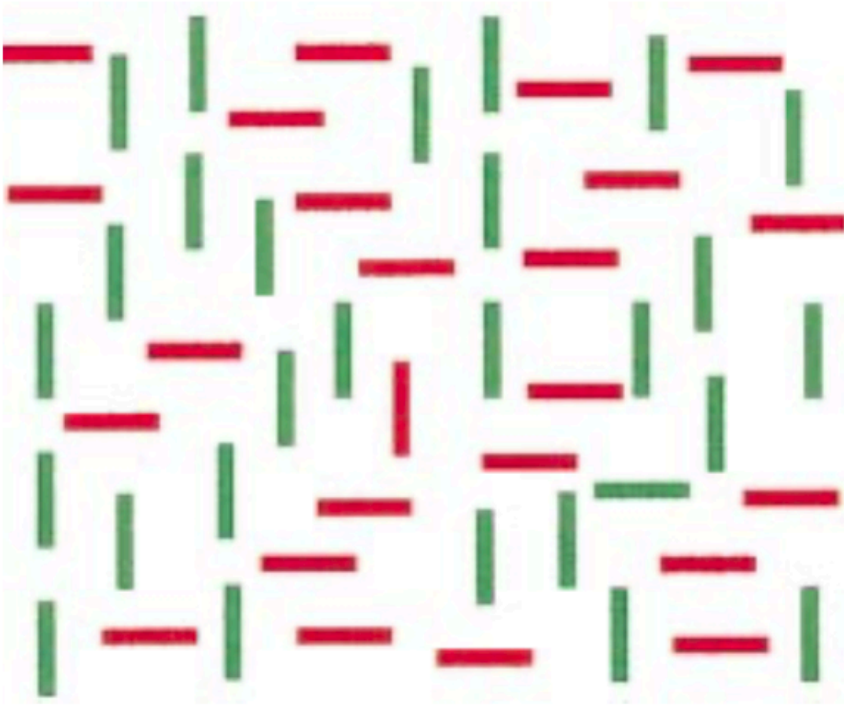
Odd *pair* of features: *serial* processing

Get ready to look for the odd pair of features!

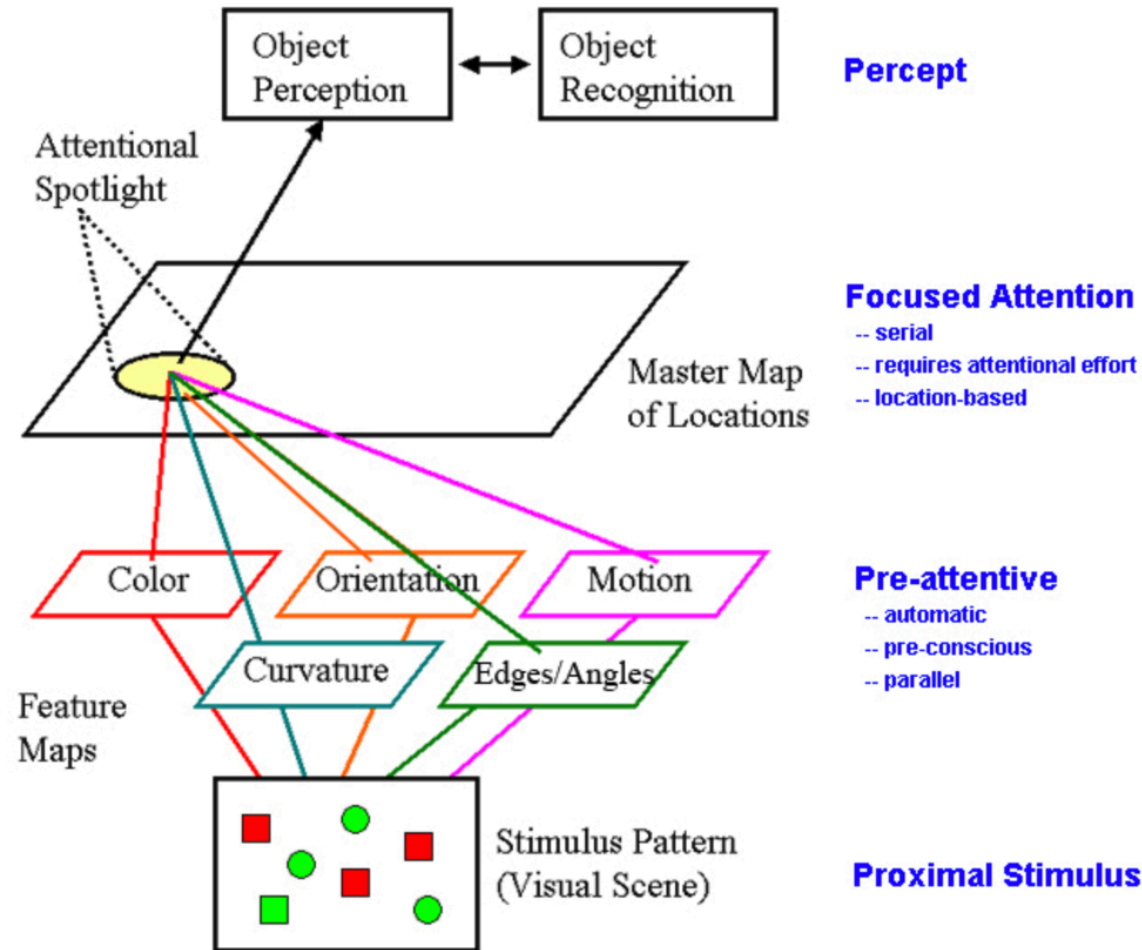




Odd feature pair: *serial* processing



Feature integration theory of Treisman



Binding features is slow because it requires attention

Pre-attentive saliency mechanisms extract odd colors.

Feature attention allows rapid selection of feature.



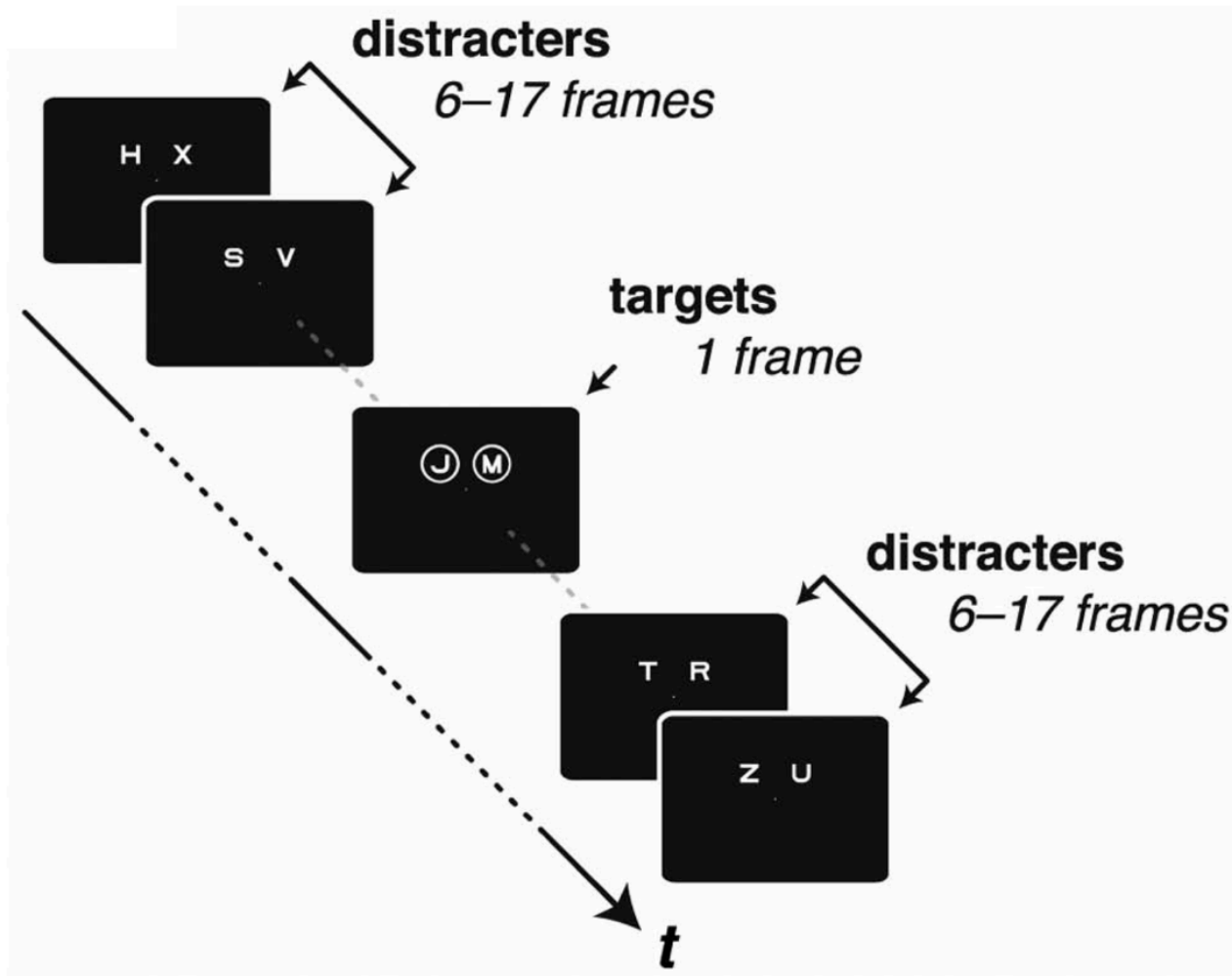
HARD QUIZZ



HARD QUIZZ

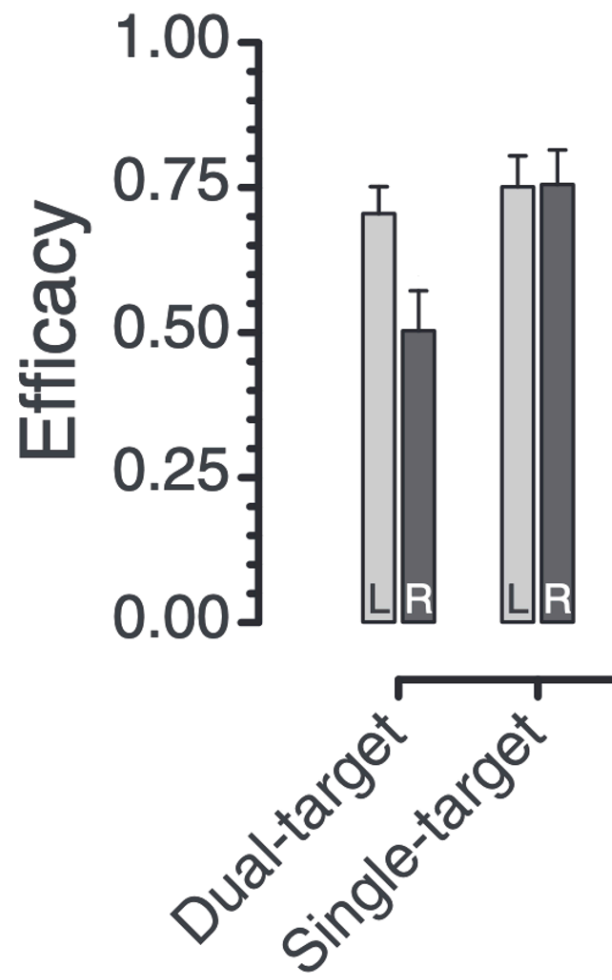


Two letters > processing capacity



Two letters > processing capacity





Feels like we see everything at once

- “phenomenal content overflows cognitive access.” - Ned Block
 - “phenomenal” consciousness
- Chapter 3.5



Feels like we see everything at once

↑ How can we reconcile this with the bottleneck?

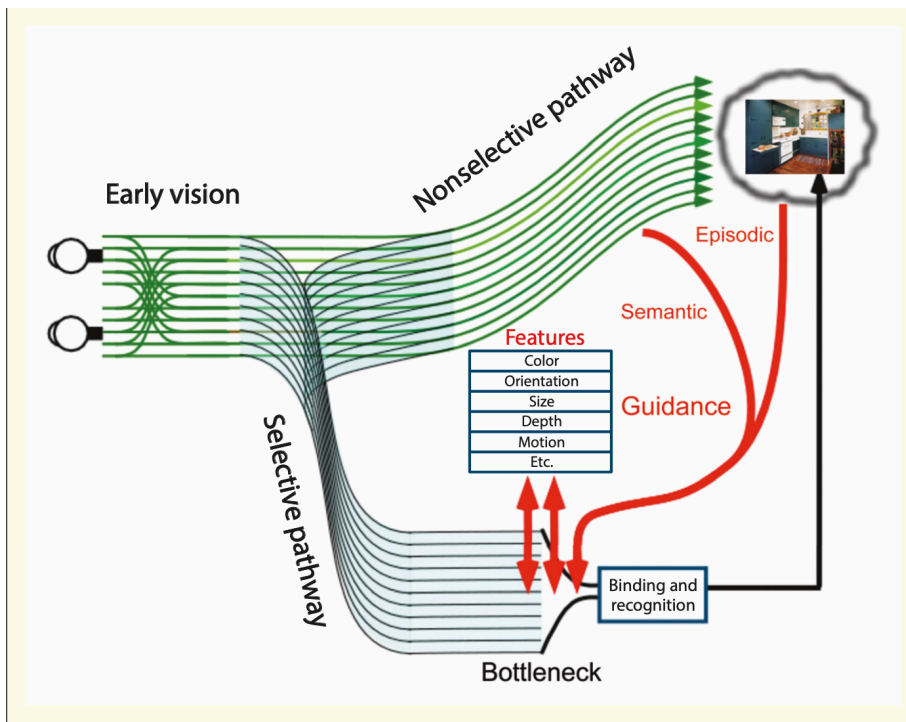
1. Attention shifts so quickly we don't notice it
 - The refrigerator light illusion.
2. Something's missing from our theories?



Wolfe et al. (2011):

there is a 'selective' path in which candidate objects must be individually selected for recognition and a '**nonselective**' path in which information can be extracted from global and/or statistical information.







PSYC2016: Attention





PSYC2016: Attention

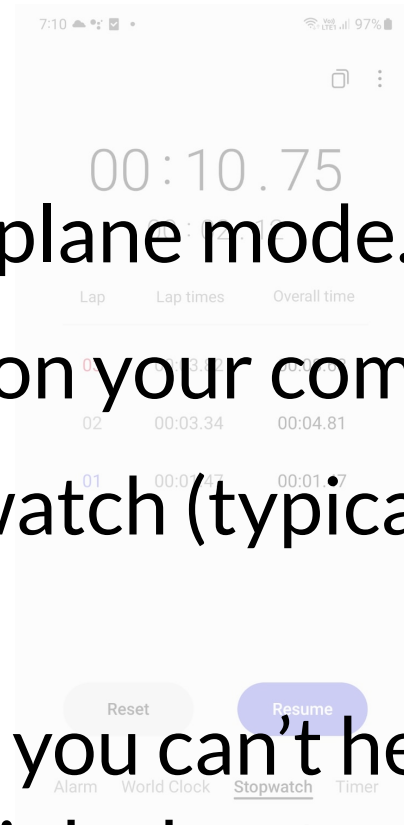


What happens when we eliminate the top 2?



Distraction experiment

- Put your phone on airplane mode.
- Turn off notifications on your computer.
- Start the phone stopwatch (typically on Android phones, in Clock).
- Press “lap” every time you can’t help yourself and you check social media, think about something other than what you’re reading, or otherwise get off task.



Let's spend X minutes silently reading Chapter 7, and 8 if you finish

Press 'lap' every time you get distracted.



menti.com

Distraction experiment results

- How many times did you press 'lap'?

Thing(s) I didn't
understand

Waiting for responses ...



menti.com

Thing(s) I didn't
understand

Waiting for responses ...





